

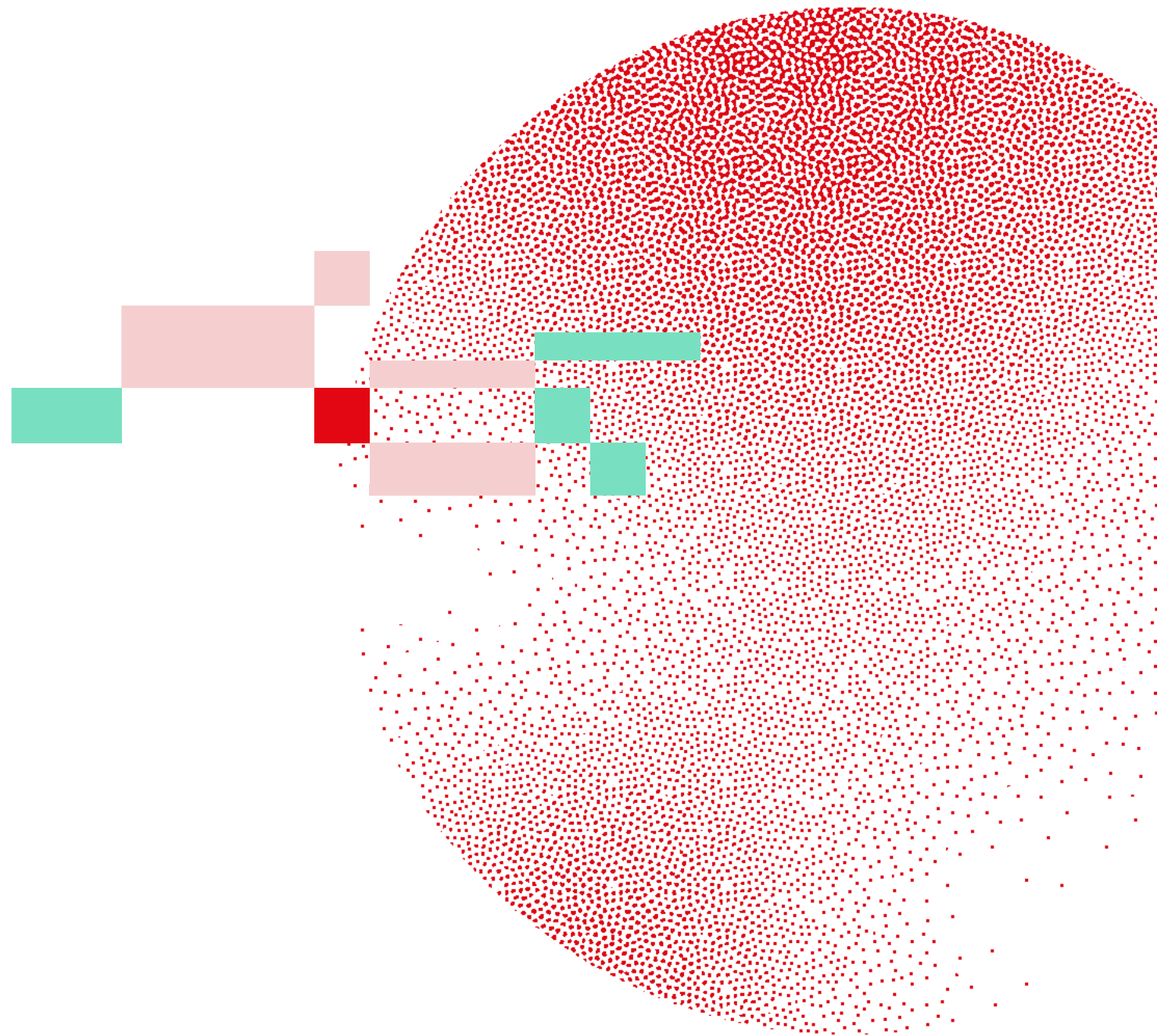


Swiss Institute of
Bioinformatics

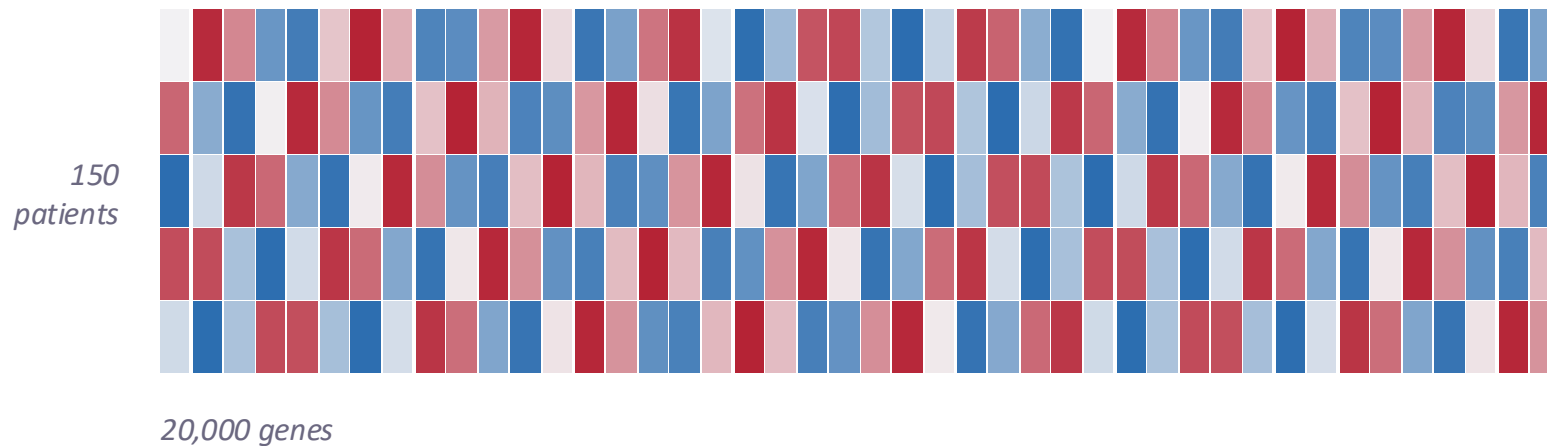
Feature selection

SIB Training Course

21/09/2026



Too many variables, too little data



$p = 20,000$

genes measured per patient

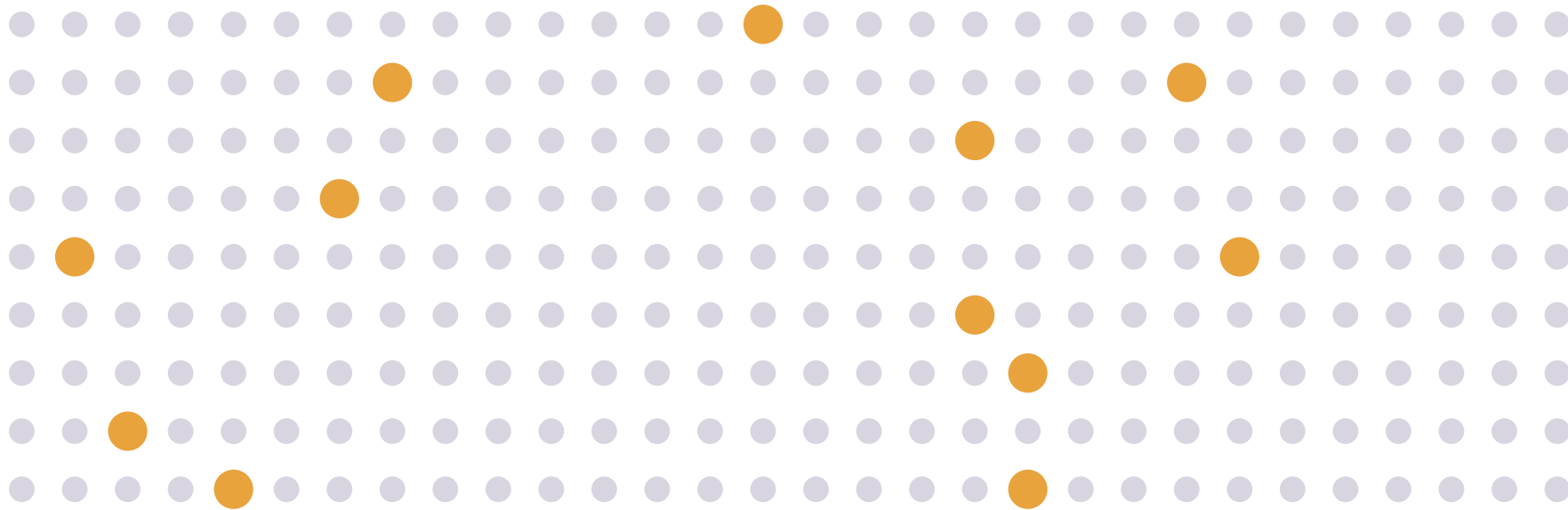
$n = 150$

patients in the cohort

133

genes for every single patient. A model can fit this data perfectly and learn nothing — and with 20,000 candidates, some genes will separate the subtypes beautifully by chance alone.

Needles in a haystack



≈ 30

informative genes

hidden among 20,000

Too strict → you miss real biology.

Too permissive → your panel is noise that will not replicate.

The roadmap for today

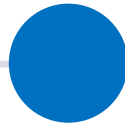
Filters

no model



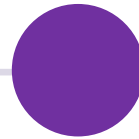
Wrapper

search



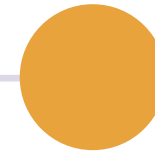
Embedded

one fit



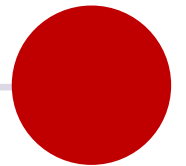
Boruta

all-relevant



Knockoffs

error control



cheap · fast · heuristic

expensive · slow · statistically rigorous

Three ways to choose genes

Filter

0

model fits

Score each gene on its own,
before any model exists.

Wrapper

many

model fits

Retrain a model again and again
inside a search over subsets.

Embedded

1

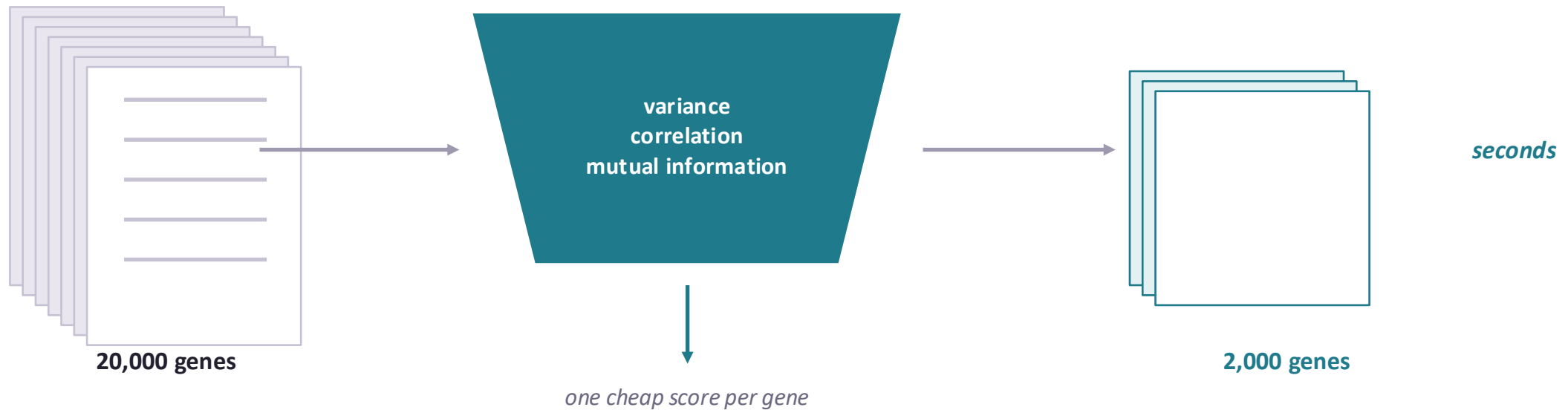
model fits

Selection happens while the
model is being fitted.

FILTER

Filters: screening the CVs

Score every gene on its own — before a single model is trained.

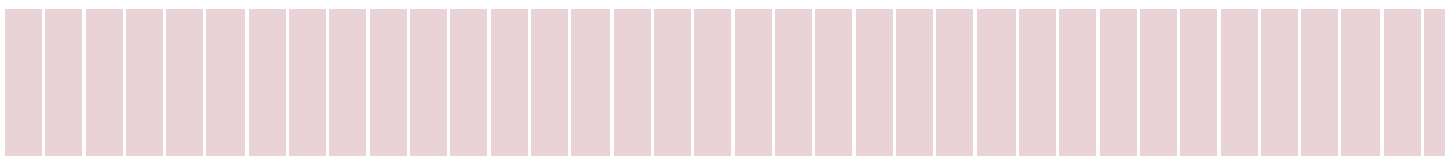


FILTER

Variance threshold

GENE-A

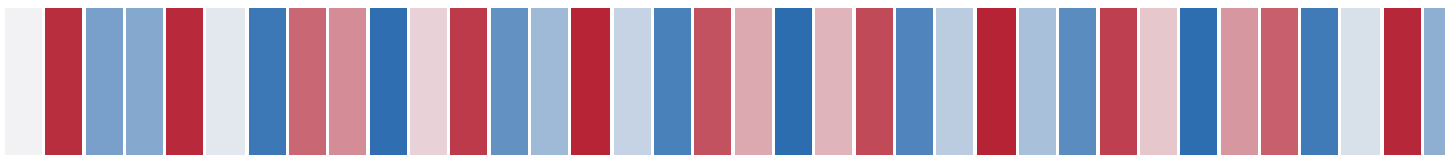
*flat across
all patients*



✗ var $\approx$ 0.001

GENE-B

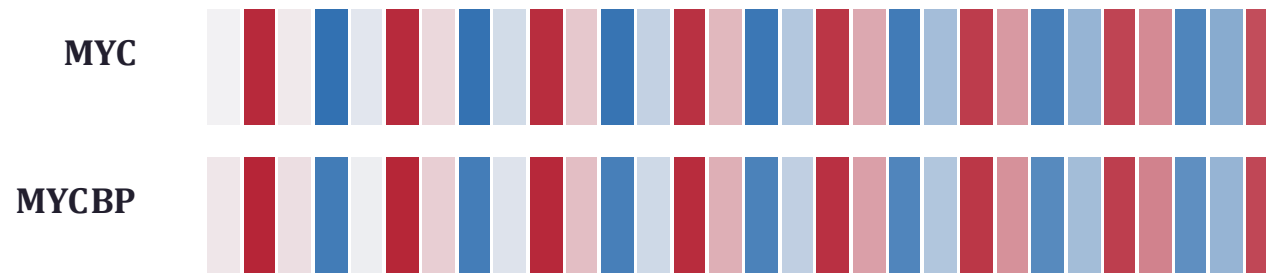
*varies with
subtype*



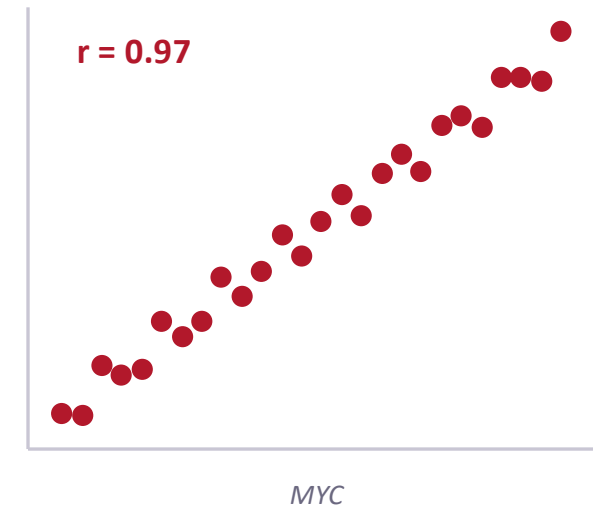
✓ var = 1.84

150 patients →

Correlation filter



the same pattern, twice

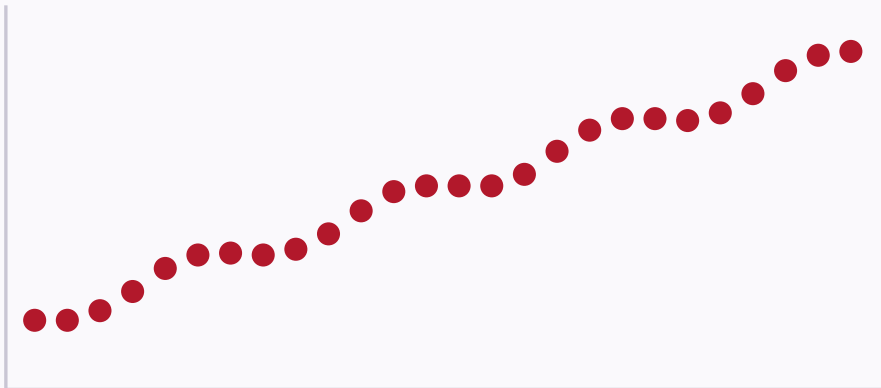


Identical twins on the same interview panel — the second one barely adds anything. But which twin you keep is essentially arbitrary, and that instability comes back to haunt you this afternoon.

Mutual information

Correlation sees straight roads. Mutual information also sees the winding ones.

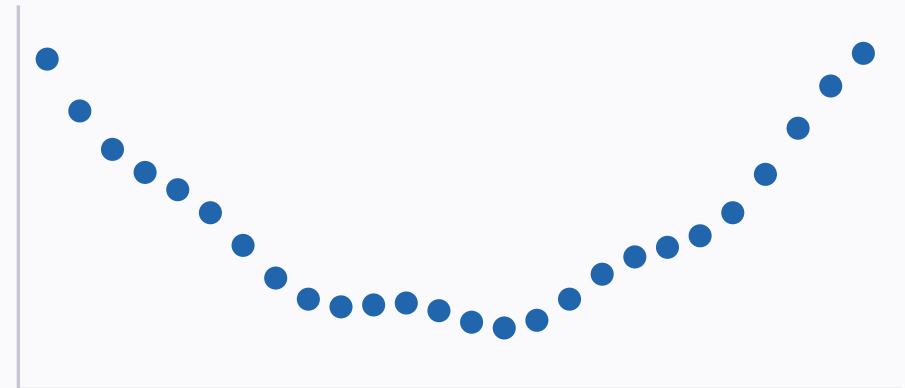
A straight road



$r = 0.91$

MI high

A winding road



$r = 0.02$

MI high



Threshold effects, bimodal genes and U-shaped dose responses are ordinary biology — and invisible to correlation. The price of MI: it needs more samples to estimate, so treat small differences as noise.

Fast, scalable but blind

Strengths

- ✓ Seconds, even on 20,000 genes
- ✓ Model-independent, reusable
- ✓ Label-free versions cannot leak

Limitations

- ✗ Blind to gene-gene interactions
- ✗ Keeps redundant co-expressed genes
- ✗ No notion of a final gene set

Use filters to make the problem tractable, then hand over to something smarter.

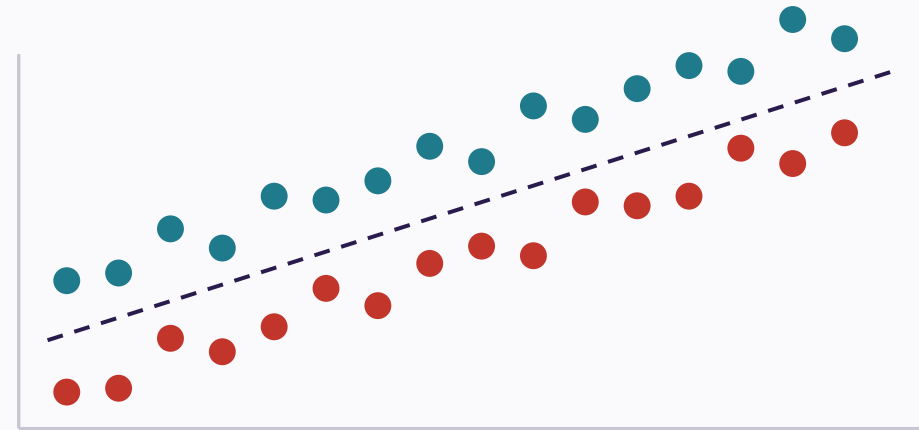
Why filters can miss biological interactions

GENE-A alone



subtypes completely overlap → filter score ≈ 0

GENE-A $\times$ GENE-B



one extra gene → clean separation

Sequential feature selection

ROUND 1

{ }

current set

try each remaining gene in its company:

+ MKI67

0.61

◀ keep

+ ESR1

0.52

+ VIM

0.41

cross-validated score of the candidate set

ROUND 2

{ MKI67 }

current set

try each remaining gene in its company:

+ ESR1

0.74

◀ keep

+ ERBB2

0.69

+ TP53

0.63

... and repeat until no candidate improves the score

a scoring rule

cross-validated, never training error

a direction

forward or backward

a stopping rule

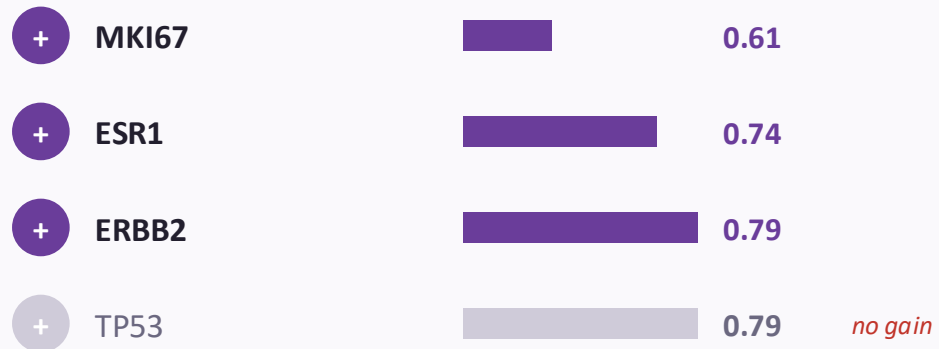
when the score stops improving

Forward versus backward selection

Forward selection

start empty, add whoever raises the score most

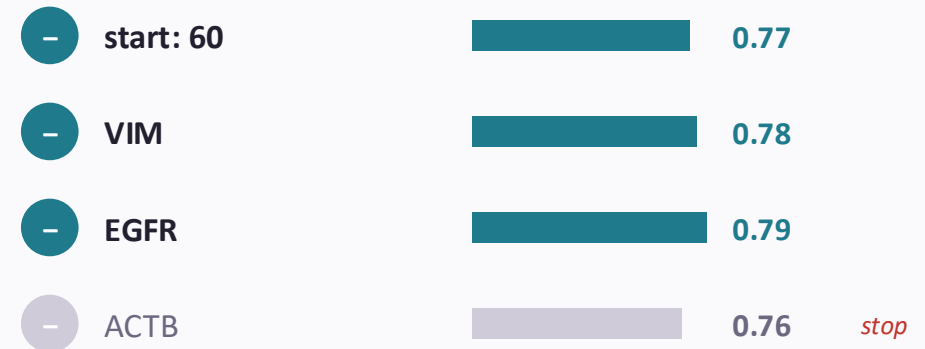
cross-validated score



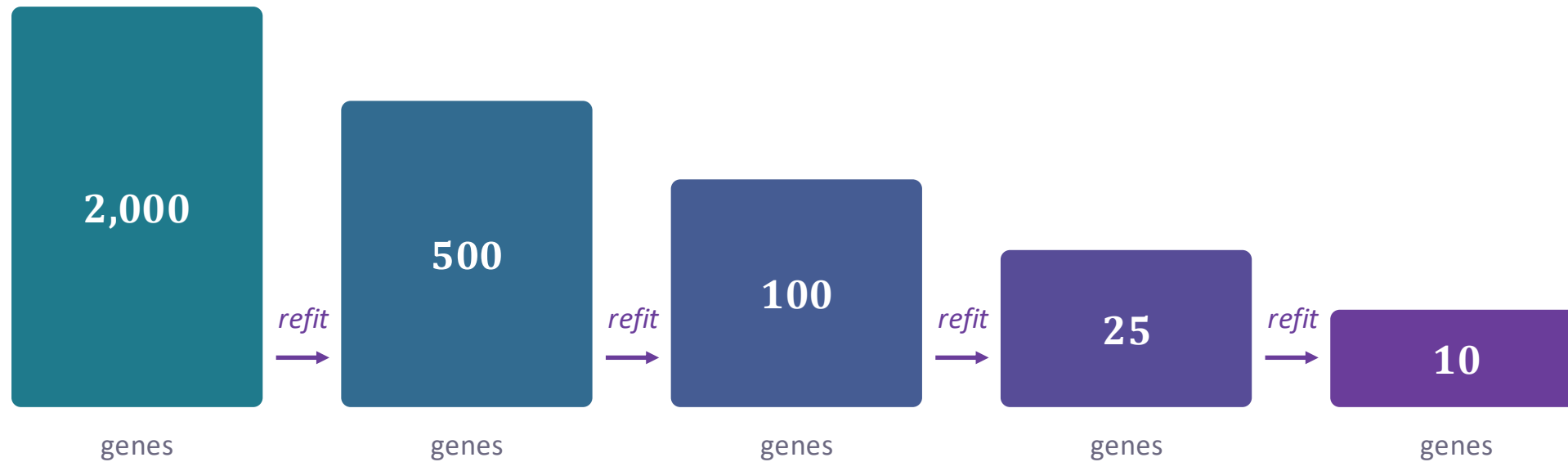
Backward elimination

start full, release whoever the model does not miss

cross-validated score



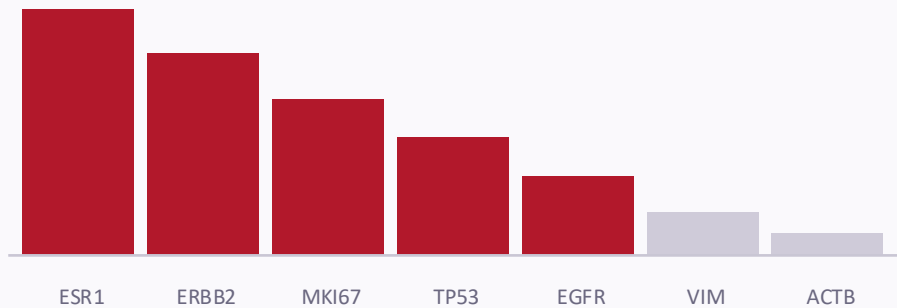
RFE: starting from the full feature set



What makes a gene the weakest?

Linear model

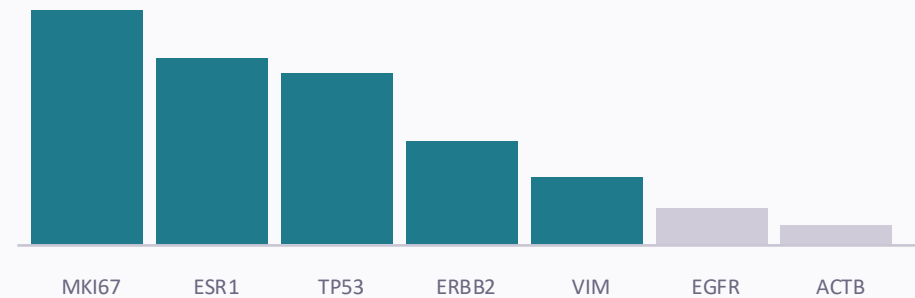
rank by | coefficient |



X dropped

Random forest

rank by impurity importance



X dropped

Accurate, but expensive

Strengths

- ✓ Sees interactions between genes
- ✓ Optimises the score you will report
- ✓ RFECV picks the subset size for you

Limitations

- ✗ Hundreds of model fits, nested in CV
- ✗ A full sequential search costs far more
- ✗ Minimal-optimal: drops redundant biology

Filter first, then let the wrapper work on what survives.

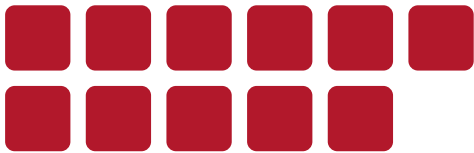
LASSO: every gene costs money

increasing penalty $\lambda \rightarrow$

small λ



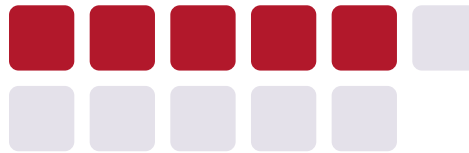
1,200 genes kept



tuned λ



80 genes kept



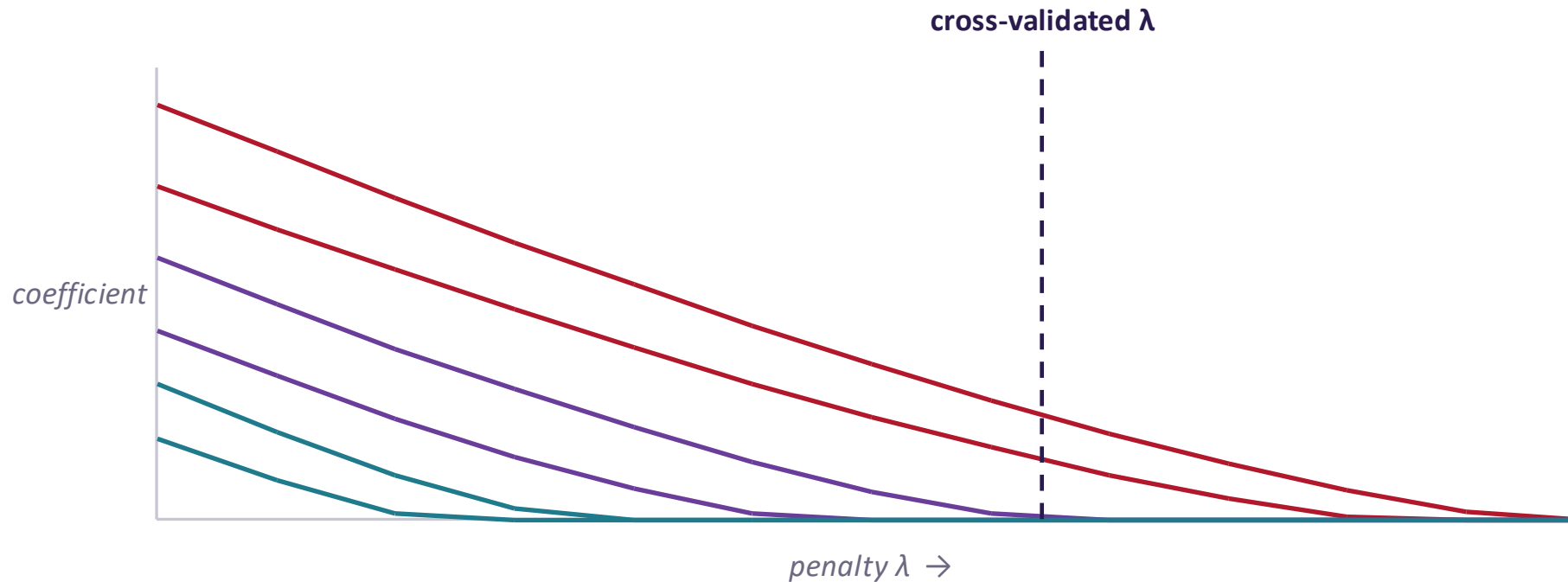
large λ



12 genes kept



Coefficients shrink to exactly zero

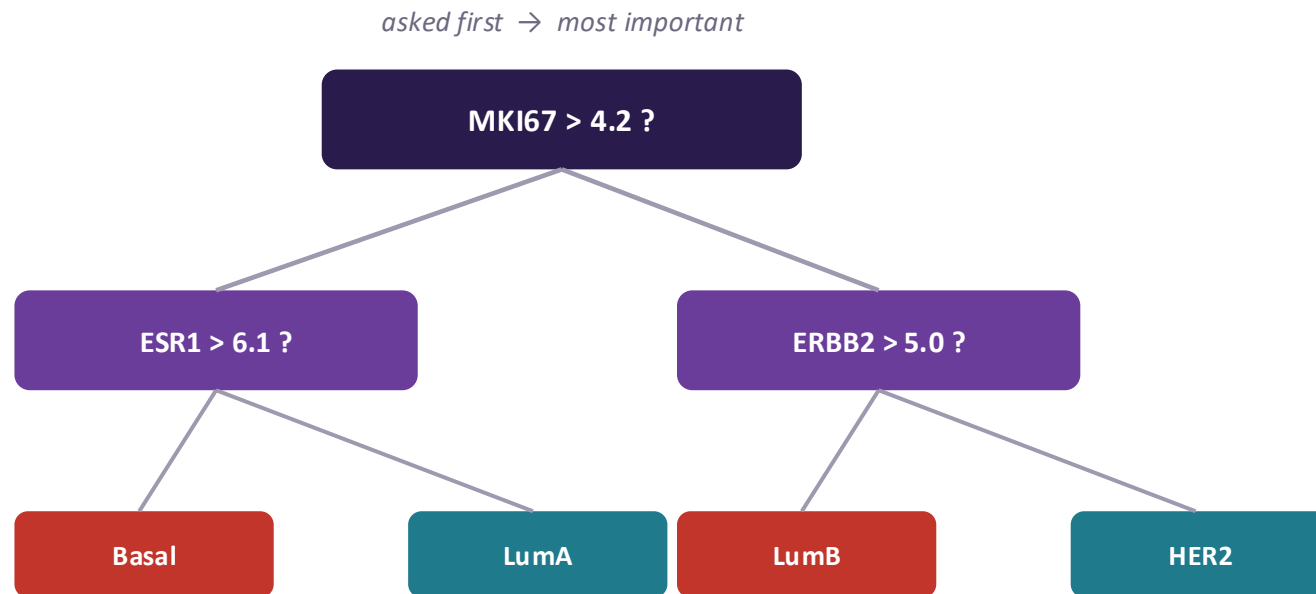


80

genes with a
non-zero coefficient

*Ridge would keep
all 2,000, just smaller.*

Trees: the Twenty Questions game



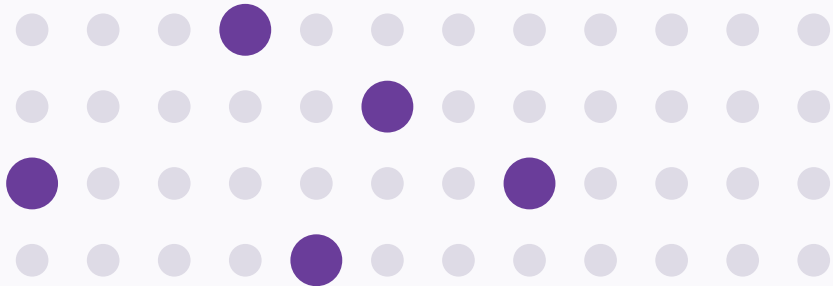
LASSO or trees?

	LASSO	Trees / forests
Model form	Linear, additive	Non-linear, interaction-aware
Correlated genes	Keeps one, zeroes the rest	Splits credit across the group
Typical output	10–100 genes, hard zeros	A score for every gene

Minimal-optimal versus all-relevant

Minimal-optimal

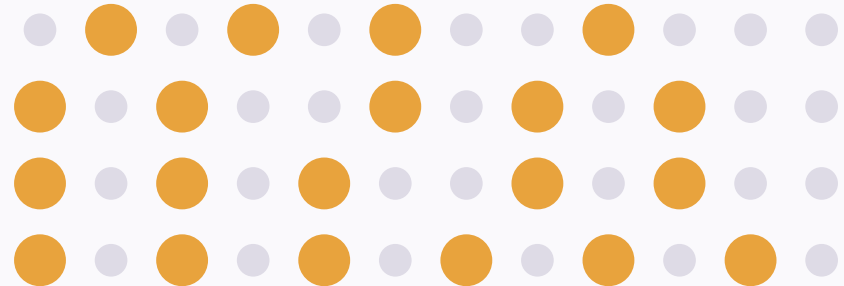
the team sheet: 5 genes that predict best



RFE · LASSO

All-relevant

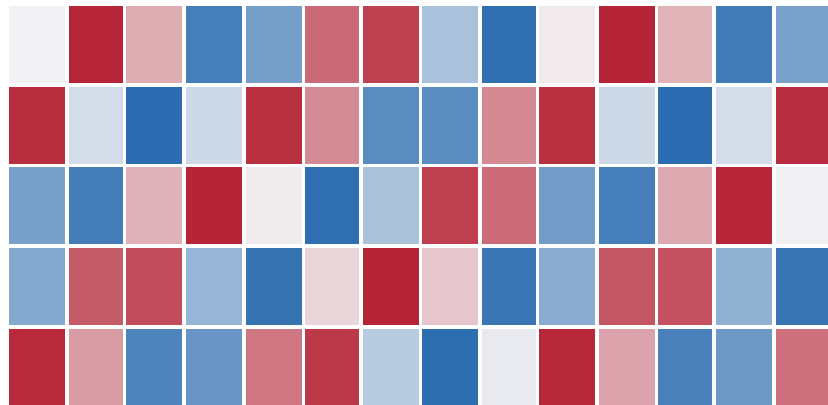
the whole club: every gene carrying signal



Boruta

Shadow features

Real genes

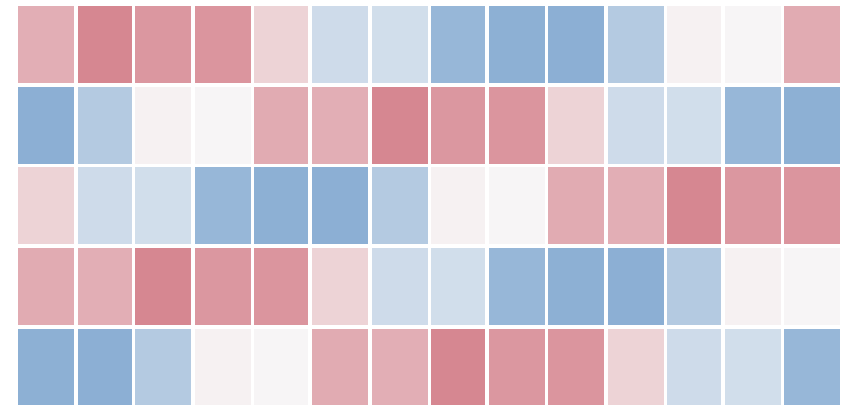


carry signal about subtype

shuffle each column



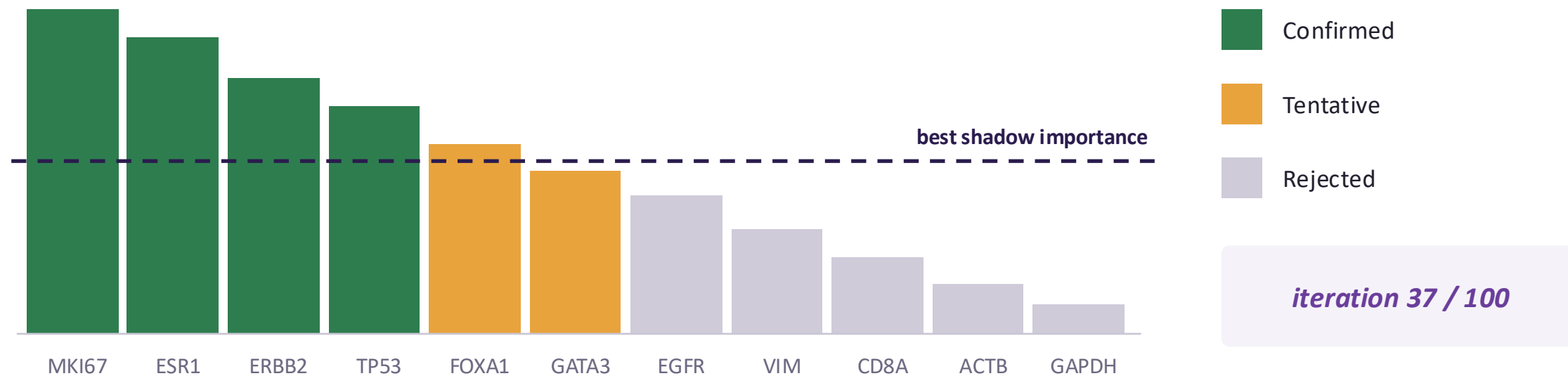
Shadow genes



same distribution, signal destroyed

BORUTA

Beat your shadow — a hundred times



Thorough, slow, generous

Strengths

- ✓ Returns all relevant genes, not a minimum
- ✓ The cut-off comes from your own data
- ✓ Tentative class is honest about doubt

Limitations

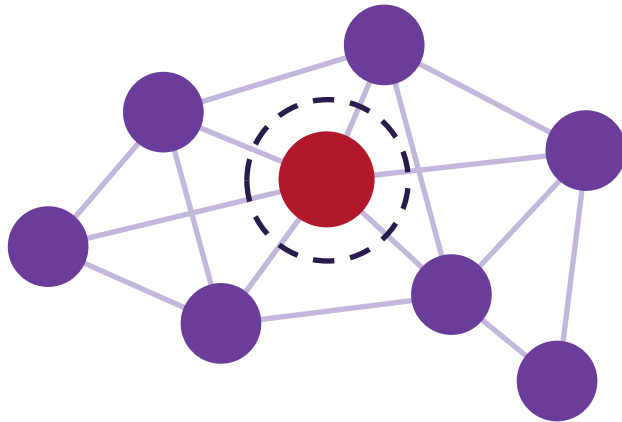
- ✗ Hundreds of forests on a doubled matrix
- ✗ No formal false-discovery guarantee
- ✗ Confirms whole co-expressed modules

Ask Boruta what is relevant. Do not ask it for a small panel.

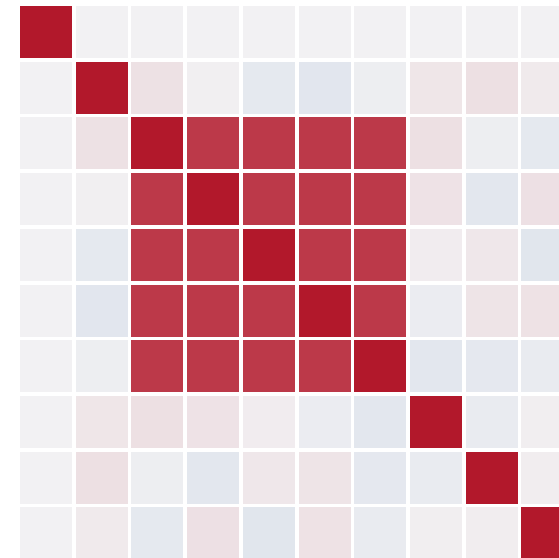
When thirty genes move as one

With co-expressed genes, most methods keep one member almost at random.

a co-expression module ($r > 0.8$)



the one LASSO happened to keep



gene–gene correlation

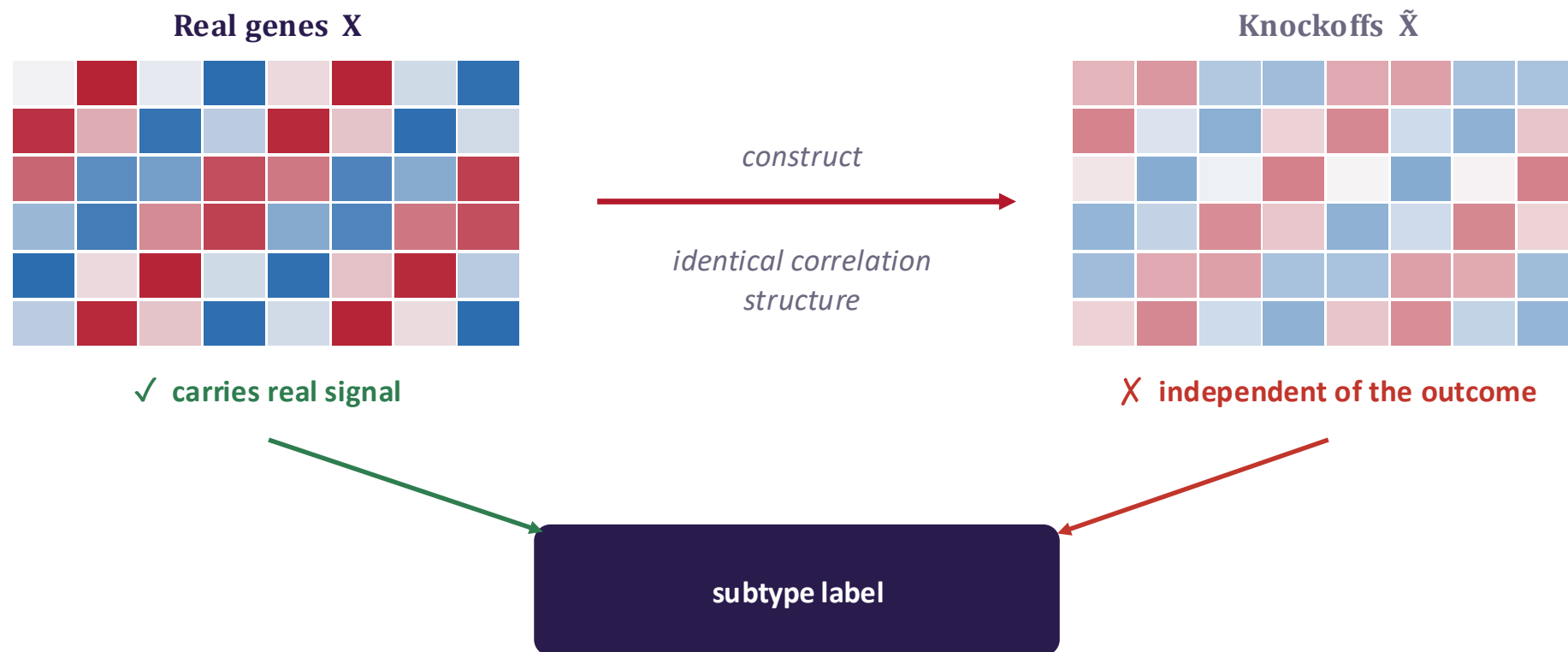
LASSO
keeps 1

Boruta
confirms 30

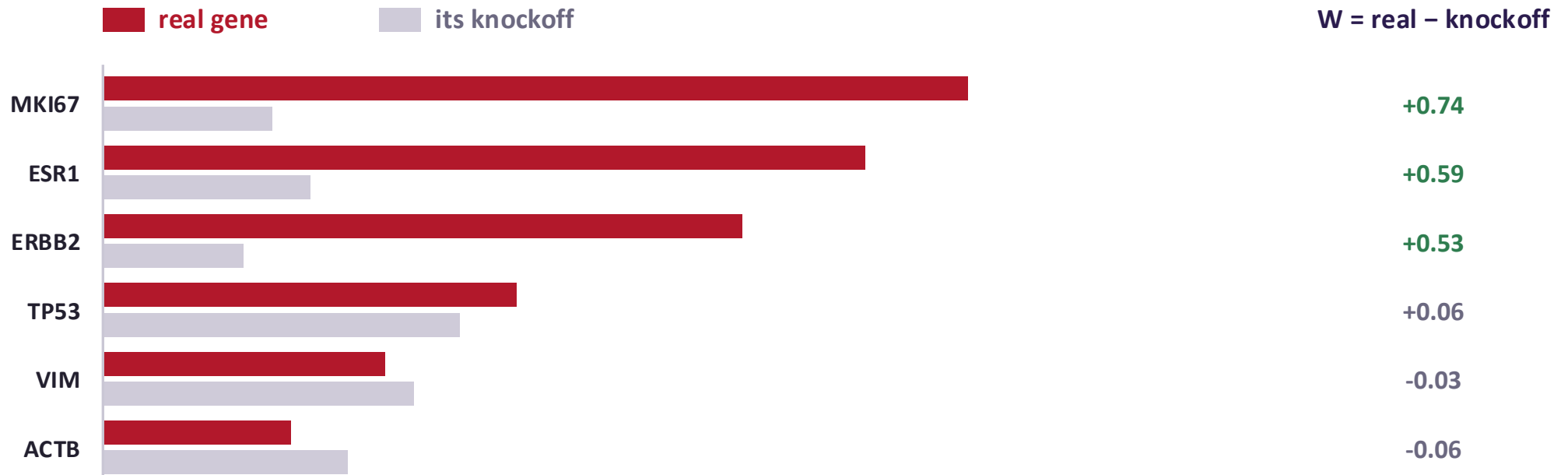
*neither tells you
how often it
is wrong*

● *Same module, two methods, two very different answers — and no error rate attached to either.*

Every gene gets a decoy



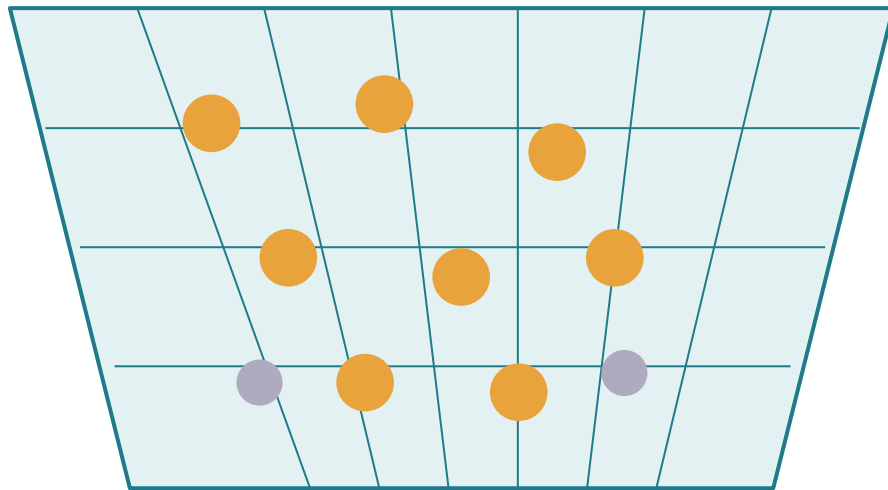
Real against decoy



Null gene: real and decoy are exchangeable → W is a coin flip.

Real signal: the gene beats its decoy systematically → W is large and positive.

Choosing your net



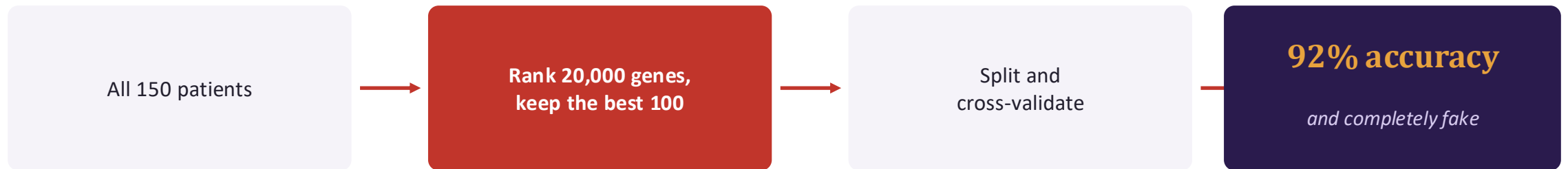
8 real discoveries · 2 false ones

target FDR	genes selected	expected false
q = 0.05	34 genes	≈ 2 false
q = 0.10	61 genes	≈ 6 false
q = 0.20	112 genes	≈ 22 false

*A finer mesh catches fewer fish and almost no debris.
A coarser mesh catches more of both.*

How to fool yourself

Select genes on all your data, and your accuracy becomes fiction.



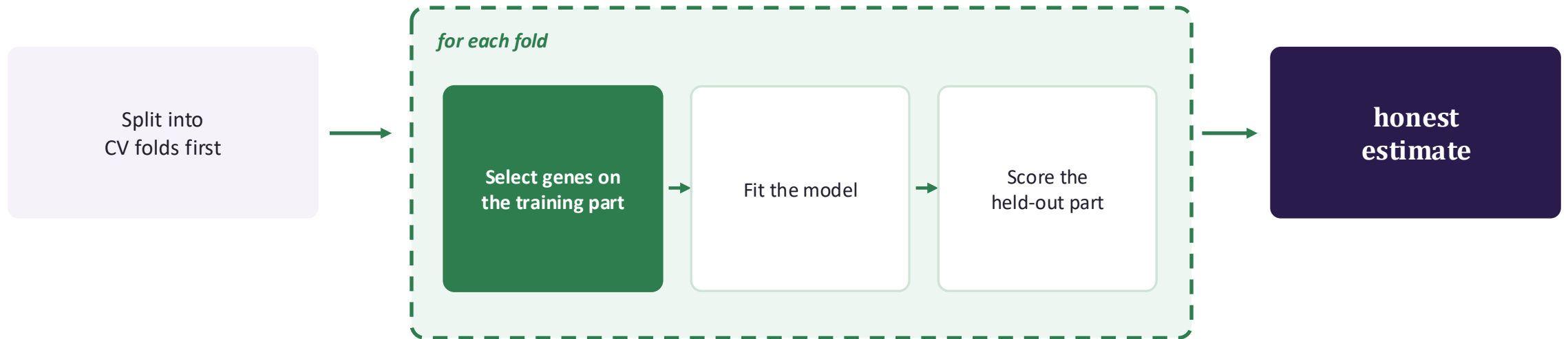
Every held-out fold was already used when the 100 genes were chosen. Shuffle the subtype labels at random and this same procedure still reports high accuracy — with 20,000 genes and 150 patients, you can always fit the noise.



A student who saw the exam paper the night before will always score well — and you learn nothing about what they know.

Selection belongs inside the fold

Every fold must choose its own genes, using only its own training data.

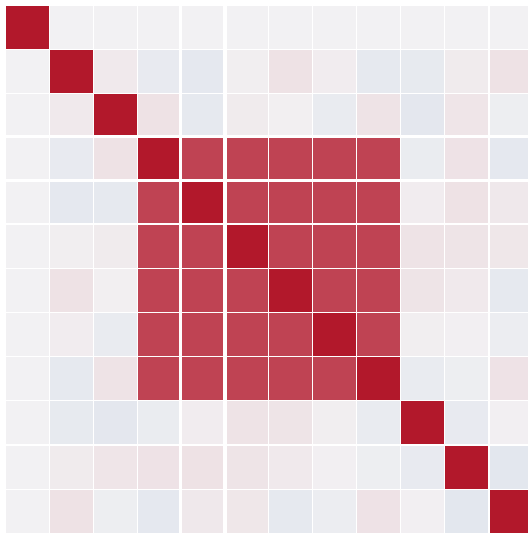


```
pipe = Pipeline([('select', SelectKBest(...)), ('clf', LogisticRegression())]); cross_val_score(pipe, X, y)
```

Different folds will select different genes. That is not a bug — it is the honest measure of how unstable selection is at $n = 150$.

Pathways move together

Correlated genes do not make your model wrong — they make your gene list arbitrary.



*one co-expressed module,
six interchangeable genes*

Run 1 · all patients



top gene: CCND1

Run 2 · drop 10 patients



top gene: CDK4

Run 3 · new random seed



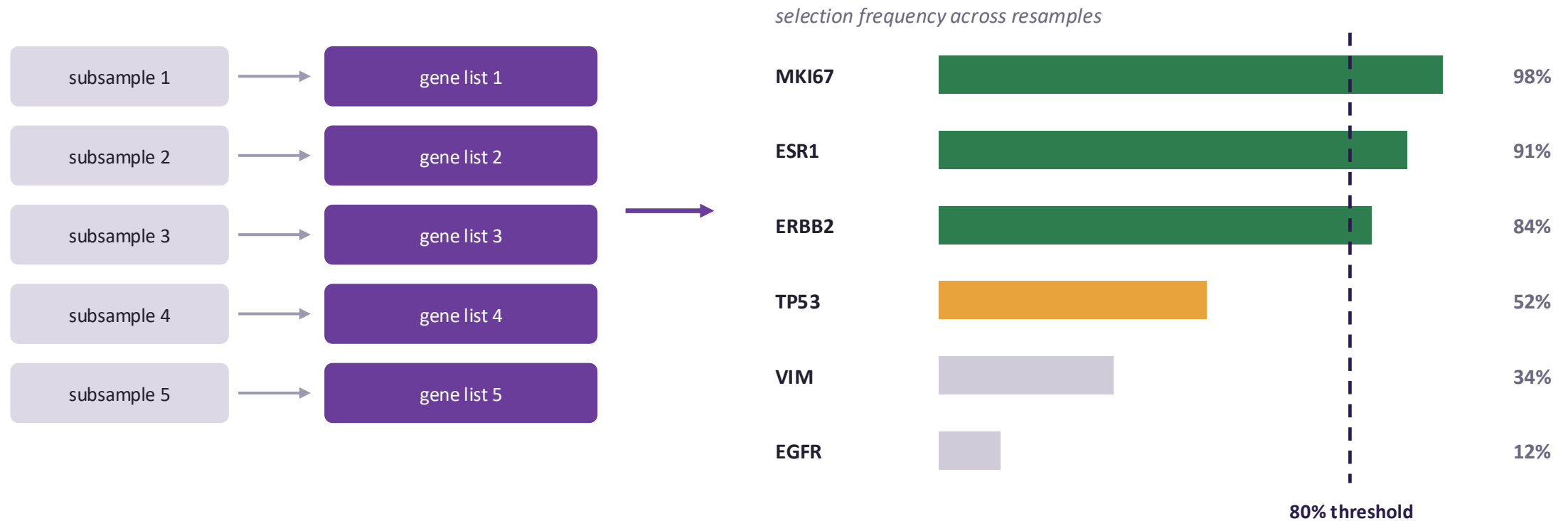
top gene: RB1

Same data, same pipeline, three different headline genes.



Report the module, not the winner. Two labs studying the same disease can publish non-overlapping signatures and both be technically correct.

Would you find the same genes tomorrow?



SHAP: explaining one patient

Global importance describes the model. SHAP describes this patient.



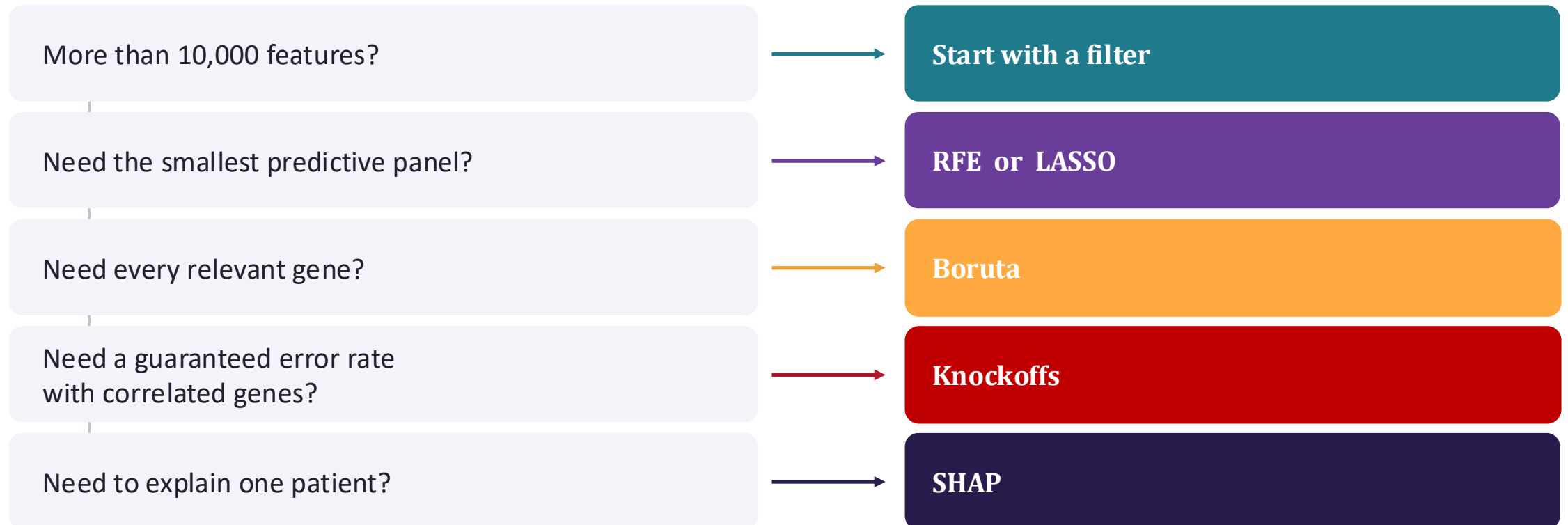
probability of subtype B →

Per-patient, not per-cohort

Which genes pushed THIS tumour toward subtype B, and by how much. Surfaces heterogeneity that a single global ranking hides — a gene decisive in a small subgroup, invisible on average.

It dissects the model, not the biology — and it is an attribution method, not a selection method.

Which method should I use?



Five ways to fool yourself

1

Selection outside cross-validation

→ Put the selector in a Pipeline

2

No stability assessment

→ Resample and report selection frequency

3

Ignoring correlation structure

→ Inspect modules, report groups not winners

4

Tuning on the data you report

→ Nested CV, or a genuinely held-out cohort

5

Top-ranked = most important biologically

→ It is the most useful to this model, on this sample

Situation to method, at a glance

High dimensionality	Filters
Best predictive subset	RFE
Sparse, interpretable model	LASSO
Biomarker discovery	Boruta
Correlated features + error control	Knockoffs

selection inside cross-validation + a stability check + honesty about correlated modules

TAKE-HOME MESSAGE

Feature selection is not about finding the best genes.

It is about finding genes that remain useful, interpretable and reproducible when new data arrive.

Before you believe your own gene list, resample the data and see whether it survives.

